

# Tone Xie

Software Engineer | Former Civil Engineer | NZ Citizen

Brooklyn, Wellington | Open to Relocate 🏠

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Website: [tx-portfolio.netlify.app](https://tx-portfolio.netlify.app) 🌐

## Work Experience

### Software Implementation Consultant | FAST Enterprises | Wellington Mar 2025 - Present

- Rebuilt the GST Groups system for New Zealand IRD using .NET and SQL Server, improving tax processing for 13,000+ businesses by automating manual workflows and enforcing compliance.
- Designed dynamic period management and processing logic for GST filing frequency changes, ensuring data integrity and robust handling of future periods and financial transactions.
- Optimized a critical SQL query for employer account attributes, reducing execution time by 70% and resolving performance issues impacting production.
- Acted as a technical consultant to IRD stakeholders, regularly presenting demos and translating tax policy into scalable software solutions under strict legislative timelines.
- Contributed to a platform that automatically issued 3.6M income tax assessments, disbursed \$3.4B in family support, and helped reduce IRD operational costs by nearly 50% (2025 figures).

### Software Development Intern | Tellus | Remote Nov 2024 - Feb 2025

- Contributed significantly to the development of a travel platform with features like real-time chat, a review management system, and page rendering optimization using Next.js and DexieDB.
- Built responsive, mobile-first UI components based on Figma designs, ensuring accessibility across devices.
- Collaborated closely with a remote team using Agile methodologies, leveraging tools such as Slack, Trello, and GitHub for communication, task management, and version control.

### Graduate Teaching Assistant | Web Development | University of Auckland Feb 2024 - Jun 2024

- Guiding 30+ postgraduate students in mastering web development technologies as a tutor, emphasizing foundational concepts, responsive design, frontend and backend frameworks, database management, and version control systems.

### Frontend Engineer Intern | Fourier Labs | Remote Nov 2023 - Jan 2024

- Created 10 new components and enhanced functionality for existing client facing Vue.js and React websites based on Figma design specifications, contributing to the development of a blockchain smart contract service.
- Collaborated with stakeholders and designers to ensure a positive feedback loop.

### Structural Engineer | Harris Foster Consulting Group | Auckland Mar 2022 - Apr 2023

- Effectively managed and coordinated work on hundreds of different projects from terrace housing and 3-story standalone homes, to housing developments up to 17 blocks at a time, to issuing CANs addressing on-site issues.
- Consistently demonstrated exceptional time management skills and work efficiency, successfully managing up to 20 concurrent projects with a strong track record of meeting deadlines and delivering within budget.

### Graduate Structural Engineer | TMC Consulting Engineers | Auckland Jun 2021 - Feb 2022

- Completed over 60 projects, including structural calculations, CAD drawings, and preparation of BC documents.

### Intern → Graduate Engineer | MEC Consultant Engineering | Auckland Dec 2019 - Feb 2021

## Skills

.NET SQL C# Java JavaScript MongoDB ExpressJs React NodeJs Svelte  
VueJs HTML CSS Figma Tailwind GitHub Python SAP Build English Native  
Mandarin Fluent AutoCAD Bluebeam SpaceGass SAP2000 LeanIX  
Microsoft Office Suite Consulting Collaboration Problem Solving

# Projects

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## Personal Portfolio Website | Website: tx-portfolio.netlify.app

- Crafted a dynamic portfolio website using React and Tailwind showcasing my projects and experience.

## AWS 3-Tier Architecture Practice Project

- Built a 3-tier web app architecture on AWS as part of a school project, using EC2 for a React frontend, Node.js backend behind an Application Load Balancer, and RDS MySQL for data storage.
- Configured NGINX as a reverse proxy to manage CORS and secure traffic between frontend and backend.
- Simulated high availability with load-balanced app servers and a multi-AZ RDS setup for fault tolerance and data reliability.

## Lecturmo | Full-Stack Student Course Review Application

- Led development and engineering of a mobile-first, MERN stack application facilitating student course reviews.
- Integrated dynamic QR code system for real-time attendance verification, enhancing review credibility.
- Leveraged OpenAI API to implement AI-driven summaries for concise and informative course reviews.
- Implemented automated email verification to confirm user attendance to UoA, further enhancing user credibility.

## Full-Stack Blog Application

- Built a secure bespoke authentication system with hashed and salted passwords, session management, and dynamic client-side verification, along with a comprehensive notification system.
- Implemented server-side rendering with Handlebars.js, using the DAO model with SQLite, and reviewed over 100 pull requests for quality and consistency.

## Space Battle | Two-Player Spaceship Game | Pygame

- Engineered a real-time multiplayer game using Python and Pygame, implementing physics-based movement, collision detection, and dual-player input handling.
- Developed a power-up system with 5 distinct boosters (Health, Rocket, Sniper, Cannon, Sonic) featuring unique visual effects and temporary gameplay modifications.
- Implemented game state management including health tracking, bullet lifetime systems, and win condition detection with collision logic.

# Education

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## Masters of Information Technology | University of Auckland | First Class Jun 2023 - Feb 2025

- **Clubs:** Web Development and Consulting Club | Software Engineering Students Association | Developers Society | Tennis Club
- **Courses:** Advanced Database Systems | Advanced Information Security | Software Tools and Techniques | Cloud Computing Architecture | Creating Maintainable Software

## Postgrad Cert. of Information Technology | University of Auckland | GPA: 9/9 Jun 2023 - Nov 2023

- **Courses:** Programming for Industry | Programming with Web Technologies

## Bachelor of Engineering Honours | University of Queensland | Brisbane Feb 2017 - Jan 2021

## International Baccalaureate Diploma | Auckland International College Jun 2014 - Jul 2016

# Awards

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## People's Choice Award | AI Forum NZ | AI for the Environment Hackathon 2024

- Recognized for prototyping an innovative biodiversity protection solution using AI to predict stoat behavior and optimize trap placement based on historical and environmental data.

## Most Google APIs Used Award | Google Developer Student Clubs Hackathon 2024

- Awarded for integrating 7 Google APIs into a health tracking application featuring a virtual pet that reflects user health metrics through React, Express, Supabase, Prisma, and OAuth authentication.